(5) $k = 5,\ a = 0,\ b = 4$

a. $L_1 = \lim_{x \to 5} \frac{(x^2 - 5^2)}{(x - 5)}$

$\lim_{x \to 5} = \frac{\cancel{(x-5)}(x+5)}{\cancel{(x-5)}} = (x+5) = (5+5) = 10$ //

b. (+) MATLAB

$L_2 = \lim_{x \to 0} \frac{\sqrt{x + (b)^2} - b}{x}$ $\quad A = \sqrt{x+b^2} - b$

$B = \sqrt{x+b^2} + b$

$\lim_{x \to 0} = \frac{\sqrt{x+b^2} - b}{x} \cdot \frac{\sqrt{x+b^2} + b}{\sqrt{x+b^2} + b}$

$= \frac{x + b^2 - b^2}{x(\sqrt{x+b^2} + b} = \frac{1}{\sqrt{x+b^2} + b} = \frac{1}{\sqrt{b^2} + b} = \frac{1}{b+b} = \frac{1}{12}$ //

$\lim_{x \to \infty} = \frac{2x^3 + 4x + 7}{x^3 - x + 5}$

$= \frac{2 + \frac{4}{x^2} + \frac{7}{x^3}}{1 - \frac{x}{x^2} + \frac{5}{x^3}} = \frac{2}{1} = 2$ //

c. $L_4 = \lim_{x \to 0} \frac{\sin((a+1) \cdot x) \cdot \tan((b+2) \cdot x)}{x^2} = \lim_{x \to 0} \frac{\sin(x) \cdot \tan(bx)}{x^2}$

$= \left(\frac{\sin x}{x} \cdot \frac{\tan(bx)}{x}\right) \rightsquigarrow$ Pisahkan lagi $= \frac{\tan(bx)}{x} = \frac{\tan(bx)}{bx} \cdot b$

$= \lim_{x \to 0} \left(\frac{\sin x}{x} \cdot \frac{\tan(bx)}{bx} \cdot b\right) = 1 \cdot 1 \cdot b = b$ //

absah secara matematis karena mengikuti teorema limit produk (perkalian)

d. $R(n) = \dfrac{(100.n+k)}{(n+(a+b+1)}$, $k = 5$, $a = 0$, $b = 4$

$$R(n) = \frac{100n+5}{n+5}$$

$$\lim_{n\to\infty} R(n) = \lim_{n\to\infty} \frac{100n+5}{n+5} = \frac{100+\frac{5}{n}}{1+\frac{5}{n}} = \frac{100}{1} = 100$$

berarti dalam jangka waktu sangat panjang, pendapatan bulanan dari pelanggan reguler ini stabil, pendapatan rata-rata stabil per bulan (steady-state revenue) yang bisa diharapkan dari pelanggan tidalah 100 (tidak tau dalam bentuk jt/ribu)

e. $\lim_{n\to\infty} \sqrt{n^2+k.n} - \sqrt{(n^2+b.n)}$

$$\lim_{n\to\infty} \sqrt{n^2+5n} - \sqrt{n^2+4n} \cdot \frac{\sqrt{n^2+5n}+\sqrt{n^2+4n}}{\sqrt{n^2+5n}+\sqrt{n^2+4n}}$$

$$L = \lim_{n\to\infty} \frac{(\sqrt{n^2+5n} - \sqrt{n^2+4n})(\sqrt{n^2+5n}+\sqrt{n^2+4n})}{\sqrt{n^2+5n}+\sqrt{n^2+4n}}$$

$$= \lim_{n\to\infty} \frac{(n^2+5n)-(n^2+4n)}{\sqrt{n^2+5n}+\sqrt{n^2+4}}$$

$$= \lim_{n\to\infty} \frac{n}{\sqrt{n^2+5n}+\sqrt{n^2+4}}$$

$$= \lim_{n\to\infty} \frac{\cancel{n}}{\cancel{n}.\sqrt{1+\frac{5}{n}} + \cancel{n}.\sqrt{1+\frac{4}{n}}} = \frac{1}{\sqrt{1+\frac{5}{n}}+\sqrt{1+\frac{4}{n}}}$$

$$= \frac{1}{\sqrt{1+0}+\sqrt{1+0}} = \frac{1}{1+1} = \frac{1}{2}$$

2 model ciu dengan bentuk akar hampir sama dapat dianggap sebagai 2 skenario promosi yang mirip pertumbuhan nya, tetapi satu sedikit lebih unggul

f.

$$c(x) = \begin{cases} \dfrac{(x^2-k^2)}{(x-k)} & , \text{ jika } x < k \\ p & , \quad x = k \\ a.x+q & , \quad x > k \end{cases}$$

F. ∘ hitung limit kiri saat $x \to k^-$

untuk $x < k$

$$c(x) = \frac{x^2-k^2}{x-k}$$

Faktorkan pembilang : $x^2-k^2 = (x-k)(x+k)$

selama $x \neq k$

$$\frac{x^2-k^2}{x-k} = \frac{(x-k)(x+k)}{x-k} = x+k$$

maka

$$\lim_{x \to k^-} c(x) = \lim_{x \to k} (x+k) = k+k = 2k$$

∘ hitung limit kanan

Untuk $x > k$

$$c(x) = ax + q$$

jadi

$$\lim_{x \to k^+} c(x) = ak + q$$

∘ syarat kontinu

☆ limit kiri = limit kanan

$$2k = ak + q \rightarrow q = 2k - ak = k(2-a)$$

☆ nilai fungsi = limit

$$c(k) = p = \lim_{x \to k} c(x) = 2k$$

jadi -- $\boxed{p = 2.k \ , \ q = k(2,a)}$

9 MATLAB